17 OCT 2026

25AIMLPCC303ESE

Seat No. : 36152

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St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

End Semester Examinations, October 2026

B.Tech.AIML/CSDS/COMP - SEMESTER III

25AIMLPCC303 - Microprocessor and Computer Architecture

Time : 1 Hour **Maximum Marks : 30**

Instructions

- All Questions are compulsory
- Assume suitable data wherever required

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		Attempt all Questions	Mrk	Course Outcome	RBT Level
1.	a	Apply the arithmetic instructions of the 8086 instruction set to perform arithmetic operations with suitable examples.	5	CO1	L3
		OR			
	b	i. Analyze the Von Neumann bottleneck and explain its effect on computer system performance.	3	CO1	L4
		ii. Define CPU and list its major components.	2	CO1	L1
2.	a	examine the principles of parallel processing and analyze the role of task parallelism and data parallelism in high-performance computing.	5	CO2	L4
		OR			
	b	i. Apply the concept of a hardwired control unit to explain how control signals are generated for executing an instruction.	3	CO2	L3
		ii. Explain how the next microinstruction is selected during microinstruction sequencing.	2	CO2	L2
3.	a	Compare the Minimum Mode and Maximum Mode operations of the 8086 microprocessor based on their features and applications.	5	CO3	L3
		OR			
	b	i. Identify the difference between the roles of CS and DS registers.	3	CO3	L4
		ii. Explain the difference between the minimum mode and maximum mode of operation.	2	CO3	L2
4.	a	Analyze and compare the architecture, functions, and applications of the 8257 DMA controller, 8259 Programmable Interrupt Controller (PIC), and 8255 Programmable Peripheral Interface (PPI).	5	CO4	L3

		OR			
	b	i. Compare the roles of the IRR and IMR in determining whether an interrupt reaches the Priority Resolver.	3	CO4	L4
		ii. Define RAM and ROM in the context of memory interfacing.	2	CO4	L1
5.	a	Explain the Virtual 8086 Mode of the 80386.	5	CO5	L2
		OR			
	b	i. Explain the role of the EFLAGS register and describe any three important flags.	3	CO5	L3
		ii. Explain the major functional units of the 80386 architecture.	2	CO5	L2
6.	a	Analyze how Hyper-Threading improves processor resource utilization and overall system performance.	5	CO6	L4
		OR			
	b	i. Compare the 8086 and 80386 microprocessors with respect to their architecture, word size, memory addressing capability, and operating modes.	3	CO6	L4
		ii. Define NetBurst microarchitecture used in Pentium 4.	2	CO6	L1

* All the Best *